Jan Škvrna

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jskvrna.github.io | jskvrna | Jan Škvrna | Google Scholar

Research Interests

3D Object Detection • Weakly Supervised & Self-Supervised Learning • 3D Reconstruction • Autonomous Driving

Education

PhD in Informatics Feb 2024 – Present
Czech Technical University in Prague
Visual Recognition Group (Prof. Jiří Matas), supervised by Dr. Lukáš Neumann. Research on weakly-supervised 3D object detection for autonomous driving.

MSc in Cybernetics and Robotics — with Honours Sep 2021 – Feb 2024
Czech Technical University in Prague
Thesis: Weakly-supervised 3D object detection using temporally consistent 2D cues. **Dean's Award** for exceptional thesis. Published at ECCV 2024.

BSc in Cybernetics and Robotics Sep 2018 – Jun 2021
Czech Technical University in Prague
Erasmus exchange at University of Southern Denmark (Sep 2022 – Feb 2023).

Publications

J. Škvrna, L. Neumann. **MonoSOWA: Scalable Monocular 3D Object Detector Without Human Annotations**. *International Conference on Computer Vision (ICCV)*, 2025. CVF

J. Škvrna, L. Neumann. **TCC-Det: Temporarily Consistent Cues for Weakly-Supervised 3D Detection**. *European Conference on Computer Vision (ECCV)*, 2024. Springer

Awards & Honors

Winning Solution — Structured Semantic 3D Reconstruction (S23DR) Challenge 2026
Part of the 3rd Urban Scene Modeling Workshop at **CVPR 2026**, Denver. $12k USD prize pool.

Winning Solution — Structured Semantic 3D Reconstruction (S23DR) Challenge 2025
Part of the 2nd Urban Scene Modeling Workshop at **CVPR 2025**, Nashville. $25k USD prize pool.

Dean's Award for Exceptional Master's Thesis 2024
Czech Technical University in Prague.

6th Best Master's Thesis on Industry 4.0 in Czech Republic 2024
Werner von Siemens Award.

Experience

Lab Instructor — Deep Learning Essentials 2025 – Present
Czech Technical University in Prague
Instructed all laboratory sessions for the Master's level Deep Learning course. Guided students through practical PyTorch implementations, architecture design, and model training.

Junior Embedded Software Developer Jun 2020 – Jun 2023
Poll s.r.o.
Developed real-time firmware in C/C++ for ARM MCUs (HVAC, high-voltage DC/DC converters). Contributed to FPGA VHDL implementations and Java-based diagnostics.

Technical Skills

Languages: Python, C/C++, LaTeX
ML / DL: PyTorch, NumPy, OpenCV, Open3D, PyTorch3D, Faiss
Tools: Git, Slurm (large-scale training), Linux

Academic Service

Reviewing: ICCV 2025
Invited Talk: Computer Vision Winter Workshop, Graz, Austria (Feb 2025) — *MonoSOWA*
Summer Schools: International Computer Vision Summer School (ICVSS), Sicily (2025); Vision and Sports, Prague (2024)